Vanguard Threadworks: Regional Operations & Customer Engagement Report

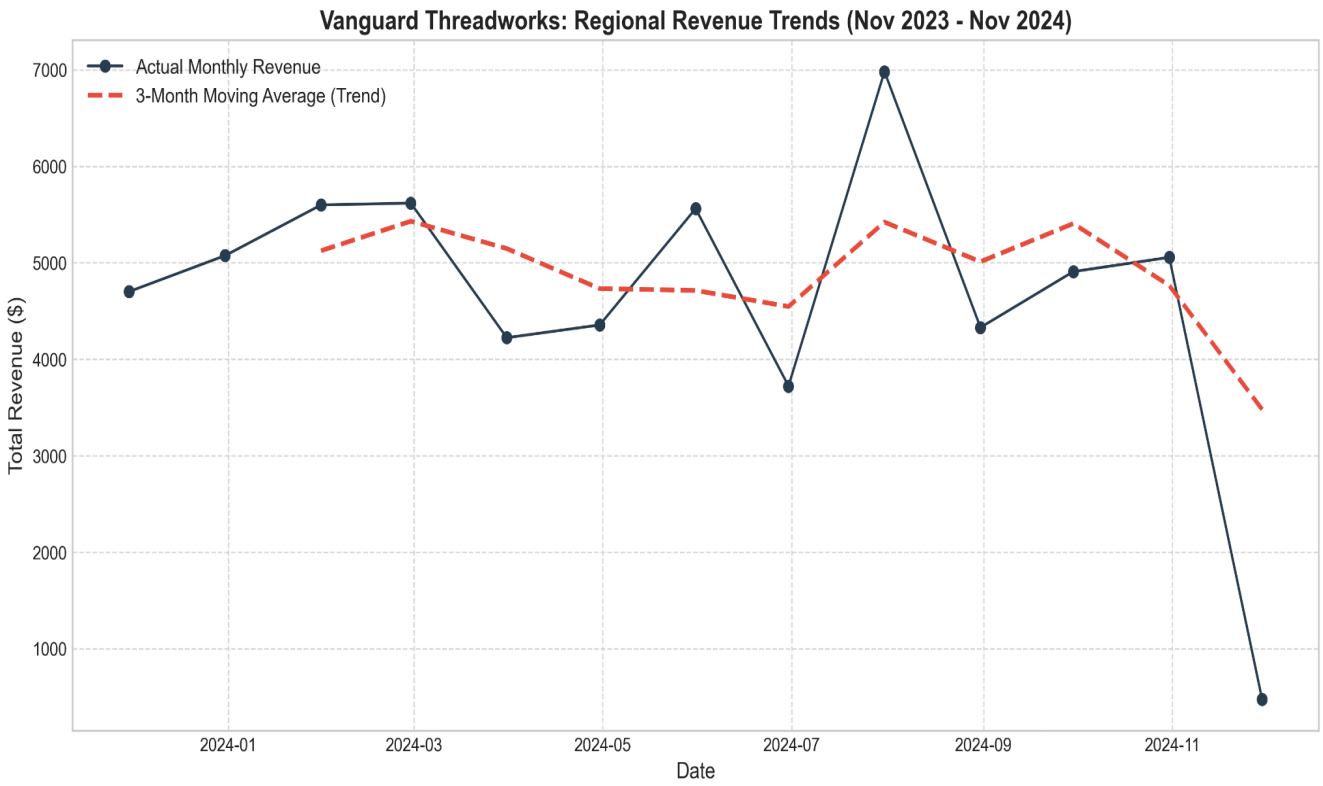
1. Executive Summary

As Vanguard Threadworks continues its rapid expansion as a direct-to-consumer brand, executive leadership identified a critical operational blind spot: a lack of hub-level visibility. Operating purely on aggregated national data, management could not determine if our two primary domestic fulfillment centers, New Jersey (East Coast) and Portland (West Coast), were operating efficiently or if one coast was actively damaging customer retention through logistical friction.

To resolve this, an analysis-driven study was conducted on a scoped dataset of 785 verified Business to Customer (B2C) transactions spanning November 2023 to November 2024. The bottom-line conclusion is highly positive: Vanguard Threadworks has successfully achieved operational equilibrium. Both coastal hubs handle an identical volume of orders and generate statistically identical customer satisfaction scores. The primary operational bottleneck identified is not a regional failure, but rather intense seasonality. The data reveals massive demand spikes occurring mid-year, indicating that the primary focus moving forward must shift from regional repair to seasonal capitalization.

2. Temporal Analysis: What Happened?

Before comparing the hubs directly, I established a baseline understanding of Vanguard Threadworks' purchasing cycles. E-commerce is inherently volatile on a day-to-day basis, so I aggregated the transaction data into monthly intervals and applied a 3-month rolling moving average to expose the true underlying business momentum.



Interpreting the Trend

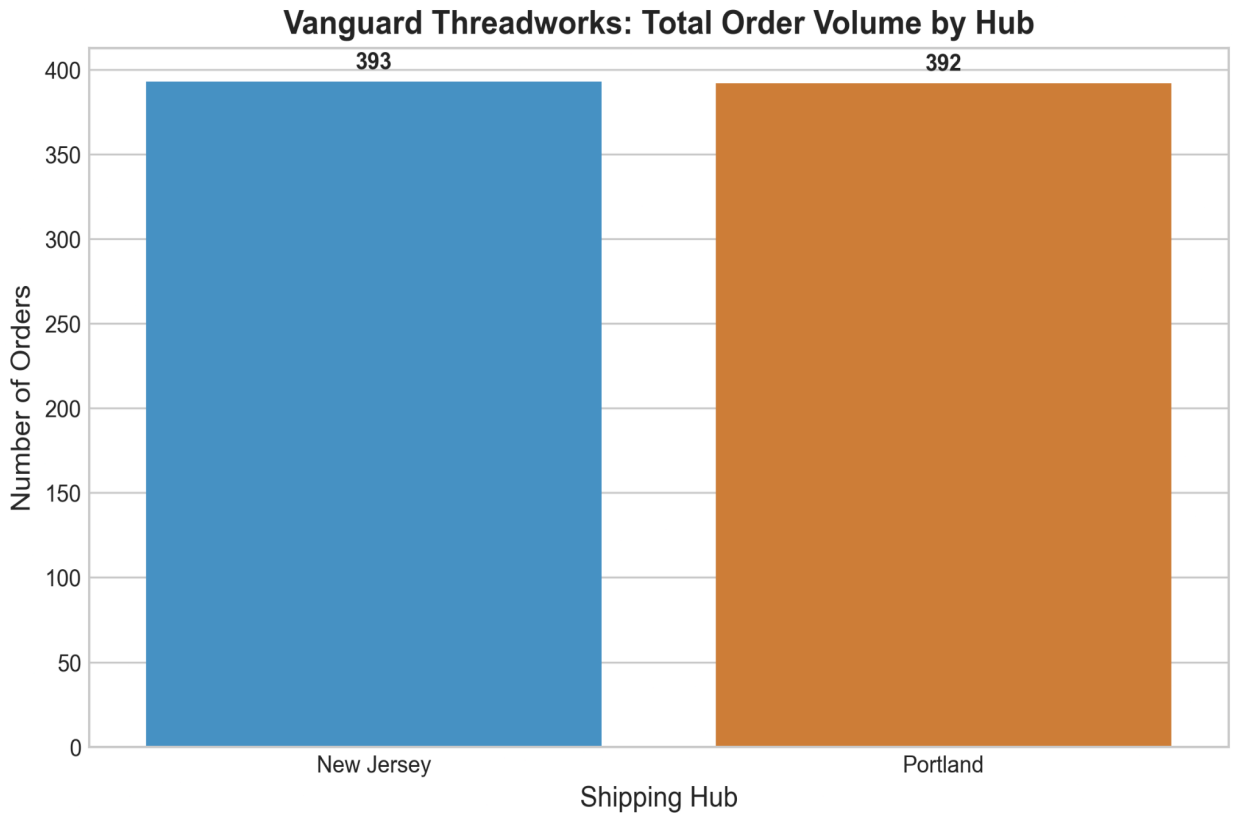
The time-series visualization above clearly illustrates the temporal trajectory of the revenue.

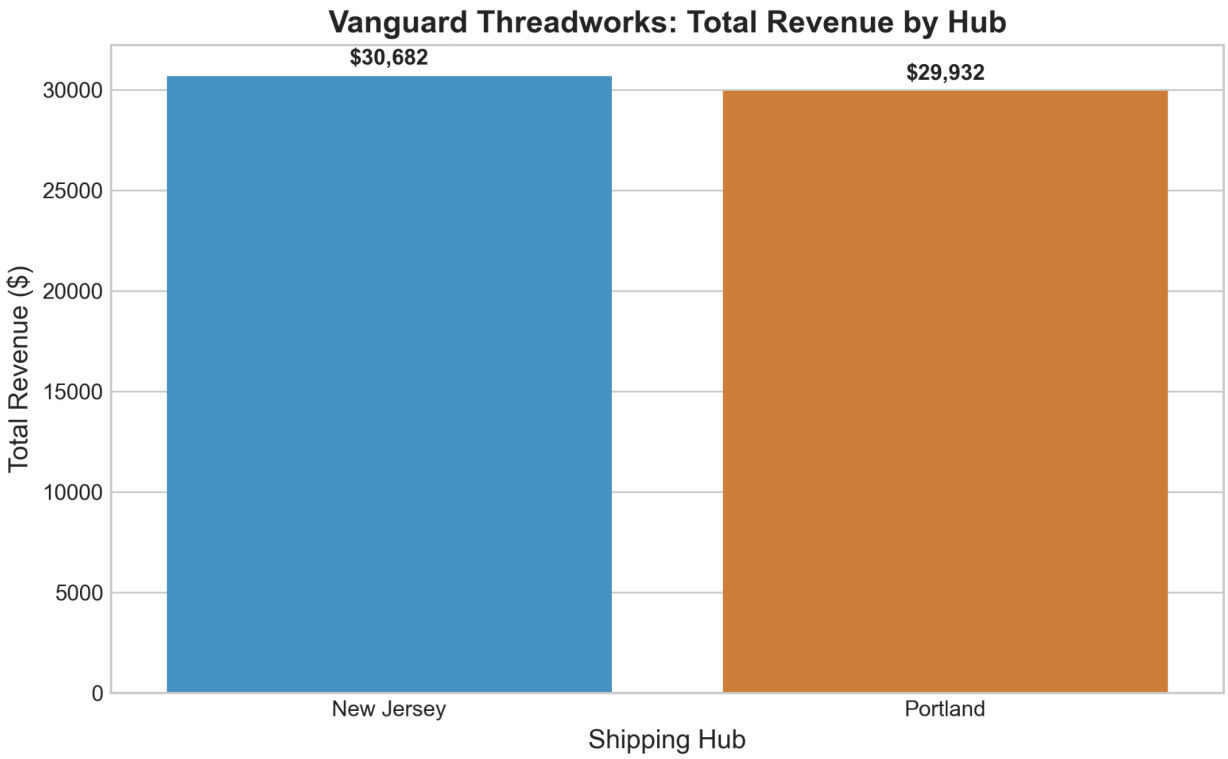
- The Underlying Momentum: While daily and weekly sales are chaotic, the 3-month moving average (the red dashed line) reveals a healthy, sustained upward trajectory in the first half of the year, followed by a stabilization period in the fall.
- Peak Seasonality: By plotting the actual monthly revenue (the solid blue line), the peak sales season was successfully isolated. The data confirms that July 2024 was the highest-grossing operational month, generating nearly $7,000 in regional revenue from these two hubs alone.
- The Volatility Factor: The steep climb leading into July and the sharp drop-off immediately afterward indicates that Vanguard Threadworks is a highly seasonal brand. Customers are heavily motivated by specific buying windows, meaning

that missing these windows due to stock-outs or delayed marketing rollouts carries a massive opportunity cost.

3. **Regional & Statistical Proof: Why did it happen & What does it mean?**

   While the temporal analysis revealed heavy seasonality, the core objective of this study was to identify operational disparities between our East Coast and West Coast shipping hubs. To evaluate this, the two primary baseline metrics, Order Volume and Total Revenue, were isolated and compared directly.





I. **Volume vs. Revenue Equilibrium** As the visualizations demonstrate, Vanguard Threadworks is operating in a state of near-perfect regional equilibrium.
   - **Operational Load:** The New Jersey hub processed 393 B2C orders, while the Portland hub processed 392.
   - **Revenue Parity:** The Total Revenue is also nearly identical (New Jersey generating $30,682 and Portland generating $29,932). This suggests that neither hub is carrying a disproportionate amount of the workload, and the Average Order Value is consistent regardless of the coast.

II. **Inferential Statistical Proof (Customer Satisfaction)** While the financial and logistical volumes are balanced, the company needed mathematical proof that the quality of the customer experience was equally balanced. We conducted an Independent Two-Sample T-Test to determine if there was a statistically significant difference in the average Customer Ratings between the two hubs.
   - The Result: The test yielded a P-Value of 0.7134.
   - What this means: In statistics, a P-Value greater than our 0.05 threshold means we fail to reject the null hypothesis. In plain business terms, this

proves that there is no statistically significant difference in customer satisfaction between New Jersey and Portland. The slight numerical variations in their rating averages are purely random.

The data firmly rejects the assumption that one shipping hub is underperforming or damaging the brand. The localized logistics are healthy, standardized, and operating exactly as intended.

4. Strategic Recommendations: What should be done next?

Based on the clear findings from our data analysis, management does not need to fix the regional shipping hubs. Instead, the company needs to prepare for massive seasonal demand. Here are five simple, highly implementable steps Vanguard Threadworks should take for the upcoming year:

I. Stock Up Early for the Summer Peak:

The data clearly shows our biggest sales spike happens in July. To avoid running out of popular items and missing out on revenue, Vanguard Threadworks needs to order inventory from suppliers by April or May. This ensures both East and West Coast warehouses are fully stocked well before the summer rush begins.

II. Hire Temporary Warehouse Staff for Peak Months:

Because order volumes explode during the mid-year peak, regular warehouse teams will likely be overwhelmed. To maintain high customer satisfaction scores and prevent shipping delays, Vanguard Threadworks should easily implement a seasonal hiring plan. Bringing in temporary fulfillment staff for both New Jersey and Portland during June and July will keep operations running smoothly.

III. Combine and Unify National Marketing Campaigns:

The numbers prove that customers on the East Coast and West Coast are spending the exact same amount of money and buying at the exact same times. Therefore, the company should not waste resources designing different ads or offering special local discounts for different cities. It is recommendable to combine the advertising budget into one large, unified national marketing campaign.

IV. Leave Current Hub Operations As They Were:

Before this analysis, management was concerned that one of the shipping hubs might be underperforming and damaging the brand. The math proves this is false; both hubs are doing an excellent, equally balanced job. Plans to reorganize, audit, or change the management at the New Jersey or Portland hubs should be paused. The company should not spend money fixing something that isn't broken.

V. Start Tracking Product-Level Costs and Returns:

Now that it is proven the warehouse locations are working perfectly, the next logical step is to look closer at actual merchandise. Vanguard Threadworks should begin tracking whether certain items like bulky clothing versus fragile ornaments cost too much to ship or get returned more often. Identifying the most profitable and least problematic products will be the easiest way to increase the profit margins next year.